

Target-Rate Least-Squares Power Allocation over Parallel Channels

Bhaskar Krishnamachari

Ming Hsieh Department of Electrical and Computer Engineering
University of Southern California, Los Angeles, CA, USA
bkrishna@usc.edu

Abstract—We study power allocation over N parallel Gaussian channels, such as OFDM subcarriers or the zero-forcing streams of a multiuser MIMO downlink, when each channel has a target spectral efficiency set by its receiver’s demand. We minimize the total squared deviation between achieved and target rates under a sum-power constraint. We prove that the optimum never overshoots a target, that the problem is equivalent to a strictly convex program, and that it has two regimes: if the budget suffices, every target is met exactly and the surplus power is left unused, unlike classical waterfilling; otherwise the budget binds. In the second regime, the optimum has a waterfilling-like closed form in which each channel has its own water level, given by the Lambert W function and proportional to the channel’s rate shortfall. The closed form extends to any ℓ_p shortfall penalty with $1 < p < \infty$, a family whose endpoints are capped waterfilling ($p = 1$) and minimax allocation ($p \rightarrow \infty$). A one-dimensional dual search with an explicit bracket finds the optimum; a compiled safeguarded Newton version converges in at most 6 iterations and runs 43 to 103 times faster than an interior-point solver for N up to 1024. On Rayleigh-faded channels at 5 dB mean SNR, the allocation serves 73.9% of demand versus 79.5% for capped waterfilling, while reducing the fraction of channels left without power from 10.1% to 3.9%.

Index Terms—Power allocation, waterfilling, parallel channels, multiuser MIMO, OFDM, Lambert W function, convex optimization.

I. INTRODUCTION

Waterfilling is the classical answer to allocating power over parallel Gaussian channels: it maximizes the sum of rates by pouring more power into stronger channels [1]–[3]. In multiuser systems, however, the parallel channels often serve *different receivers*. OFDMA subcarriers are assigned to different users, and zero-forcing precoding with unit-norm beams turns a multiuser MIMO downlink into parallel scalar channels, one per receiver stream, sharing a total power budget. Each receiver then has its own rate target derived from its demand. Rate beyond a target is power that serves no one; rate below it is demand left unmet, and how that deficit is spread across channels is how it is spread across receivers.

Existing formulations handle targets in two ways. Margin-adaptive loading treats them as hard constraints, minimizing transmit power [4] or maximizing the noise margin [5] at fixed target rates; see [6] for a survey. When the budget cannot meet every target, a choice is needed about whom to shortchange. Capped waterfilling, i.e., waterfilling with per-channel power caps [7] set at the power that meets each target, maximizes the demand served, $\sum_i \min(r_i, T_i)$. It is efficient but places

the whole deficit on the weakest channels, which receive no power at all. Fairness criteria such as proportional fairness [8], proportional rate constraints [9] and α -fairness [10] control rate *ratios*, not deviations from absolute targets.

We study a soft alternative: minimize the total squared deviation $\sum_i (\log_2(1 + a_i P_i) - T_i)^2$ under a sum-power budget. The quadratic penalty grows with the size of a shortfall, so it spreads the deficit across receivers instead of concentrating it, and, like capped waterfilling, it never spends power above a target. Our contributions are:

- **Structure.** The optimum never overshoots; the problem is equivalent to a strictly convex program; and it has a sharp two-regime structure, leaving power unused whenever every target can be met (Section III).
- **Closed form.** In the budget-limited regime, the optimum is $P_i = [L_i - 1/a_i]^+$, like waterfilling but with a channel-specific water level $L_i = 2W_0(\lambda \ln^2 2 q_i/2)/(\lambda \ln^2 2)$, where $q_i = 2^{T_i}/a_i$ and λ is a scalar dual variable; each level is proportional to the channel’s rate shortfall. The form extends to every ℓ_p shortfall penalty with $1 < p < \infty$, whose endpoints are capped waterfilling ($p = 1$) and minimax allocation ($p \rightarrow \infty$) (Section IV).
- **Algorithm.** A dual search over an explicit bracket costs N Lambert- W evaluations per iteration; with a safe-guarded Newton step it converges in at most 6 iterations (Section V).
- **Evaluation.** On Rayleigh-faded channels, the allocation gives up 2–6 points of served demand relative to capped waterfilling and leaves $2.6\text{--}4.1\times$ fewer channels without power; capped waterfilling keeps the better 5th-percentile satisfaction at 10 dB. A normalized weighting extends this behavior to mixed targets (Section VI).

Related work. Practical algorithms exist for generalized waterfilling with multiple water levels and multiple constraints [11], and mercury/waterfilling gives the optimal allocation for arbitrary input distributions [12]. The Lambert W function [13] yields closed-form power allocations in cooperative relay networks [14] and in energy-aware link adaptation [15]. Here it solves a different equation, the balance between a channel’s rate shortfall and its marginal rate. To our knowledge, the squared target-deviation objective, its Lambert- W solution, and the resulting ℓ_p family have not been treated before.

II. SYSTEM MODEL AND PROBLEM

Consider N parallel Gaussian channels with gain-to-noise ratios $a_i > 0$, known at the transmitter. Power $P_i \geq 0$ on channel i yields spectral efficiency $r_i(P_i) = \log_2(1 + a_i P_i)$. Each channel has a target $T_i > 0$ (a channel with $T_i = 0$ receives no power and is dropped). We solve

$$\min_{\mathbf{P} \geq 0} J(\mathbf{P}) = \sum_{i=1}^N (r_i(P_i) - T_i)^2 \quad \text{s.t.} \quad \sum_{i=1}^N P_i \leq P_{\text{tot}}. \quad (1)$$

Two quantities recur. The cap $\bar{P}_i = (2^{T_i} - 1)/a_i$ is the power that meets target i exactly, and

$$q_i \triangleq \bar{P}_i + \frac{1}{a_i} = \frac{2^{T_i}}{a_i} \quad (2)$$

is the value of $P_i + 1/a_i$ at which channel i reaches its target. We write $\delta_i = T_i - r_i$ for the shortfall and $c = \ln 2$.

III. STRUCTURE OF THE OPTIMUM

Lemma 1 (No overshoot). *Every optimal solution of (1) satisfies $P_i^* \leq \bar{P}_i$, i.e., $r_i(P_i^*) \leq T_i$, for all i .*

Proof. If $P_i > \bar{P}_i$, replacing P_i by $\bar{P}_i$ reduces $(r_i - T_i)^2$ to zero and keeps the budget feasible, strictly decreasing J . $\square$

Lemma 1 lets us restrict (1) to the box $\mathcal{B} = \{0 \leq P_i \leq \bar{P}_i\}$, on which $J(\mathbf{P}) = \sum_i \phi_i(P_i)$ with $\phi_i(P) = [(T_i - r_i(P))^+]^2$.

Proposition 1 (Convexity). *Each ϕ_i is convex on $[0, \infty)$ and strictly convex on $[0, \bar{P}_i]$, where*

$$\phi_i''(P) = \frac{2a_i^2}{c^2(1 + a_i P)^2} (1 + c\delta_i(P)) > 0. \quad (3)$$

Hence (1) is equivalent to the strictly convex program of minimizing $\sum_i \phi_i(P_i)$ over $\mathcal{B} \cap \{\sum_i P_i \leq P_{\text{tot}}\}$, and has a unique optimum.

Proof. $(T_i - r_i)^+$ is convex and nonnegative because r_i is concave, and $t \mapsto t^2$ is convex and nondecreasing on $t \geq 0$, so the composition ϕ_i is convex [16]. Differentiating twice on $[0, \bar{P}_i]$, with $r'_i = a_i/(c(1 + a_i P))$, gives (3). Strict convexity of the objective over a convex set gives uniqueness. $\square$

The unconstrained objective in (1) is not convex beyond $r_i = T_i + 1/c$; the no-overshoot property is what makes the problem tractable.

Theorem 1 (Two regimes). (A) Surplus. *If $P_{\text{tot}} \geq \sum_i \bar{P}_i$, then $\mathbf{P}^* = \bar{\mathbf{P}}$, $J^* = 0$, and $P_{\text{tot}} - \sum_i \bar{P}_i$ is left unused.* (B) Deficit. *If $P_{\text{tot}} < \sum_i \bar{P}_i$, then $J^* > 0$ and the budget is tight, $\sum_i P_i^* = P_{\text{tot}}$.*

Proof. (A) $\bar{\mathbf{P}}$ is feasible with $J = 0$, and by Lemma 1 and uniqueness it is the optimum. (B) Meeting every target needs $\sum_i \bar{P}_i > P_{\text{tot}}$, so $J^* > 0$ and some i has $P_i^* < \bar{P}_i$. Since $\phi'_i(P) = -2a_i\delta_i/(c(1 + a_i P)) < 0$ there, a slack budget would allow increasing P_i and lowering J . $\square$

In regime (A) the formulation departs most from waterfilling, which always spends the full budget; leaving the surplus

unused saves transmit energy, in the spirit of using spare capacity to conserve terminal energy [17]. Capped waterfilling shares this property; the two differ in regime (B).

IV. CLOSED-FORM SOLUTION

Theorem 2 (Lambert- W solution). *In regime (B), the unique optimum has the waterfilling-like form*

$$P_i^* = \left[L_i(\lambda^*) - \frac{1}{a_i} \right]^+, \quad L_i(\lambda) = \frac{2}{\lambda c^2} W_0\left(\frac{\lambda c^2 q_i}{2}\right), \quad (4)$$

where W_0 is the principal branch of the Lambert W function and $\lambda^* > 0$ is the unique root of $S(\lambda) \triangleq \sum_i P_i(\lambda) = P_{\text{tot}}$.

Proof. The convex program of Proposition 1 satisfies Slater's condition, so KKT is necessary and sufficient. With multipliers λ for the budget, μ_i for $P_i \geq 0$ and ν_i for $P_i \leq \bar{P}_i$,

$$-\frac{2a_i\delta_i}{c(1 + a_i P_i)} + \lambda - \mu_i + \nu_i = 0. \quad (5)$$

Here $\lambda > 0$: with $\lambda = 0$, every channel with $P_i < \bar{P}_i$ would need $\delta_i = 0$, forcing $\mathbf{P} = \bar{\mathbf{P}}$, which is infeasible in (B). At $P_i = \bar{P}_i$, (5) gives $\nu_i = -\lambda < 0$, so no cap binds. Active channels ($0 < P_i < \bar{P}_i$, $\mu_i = \nu_i = 0$): substituting $1 + a_i P_i = 2^{T_i - \delta_i} = a_i q_i e^{-c\delta_i}$ into (5) gives $(c\delta_i)e^{c\delta_i} = \lambda c^2 q_i/2$, whose unique nonnegative root is $c\delta_i = W_0(\lambda c^2 q_i/2)$. Then $P_i + 1/a_i = q_i e^{-c\delta_i}$, and since $W_0(z)e^{W_0(z)} = z$, this equals $L_i(\lambda)$. Inactive channels ($P_i = 0$, $\delta_i = T_i$): $\mu_i = \lambda - 2a_i T_i/c \geq 0$. Because W_0 is increasing and $W_0(cT_i e^{cT_i}) = cT_i$, the condition $\lambda \geq 2a_i T_i/c$ is equivalent to $W_0(\lambda c^2 q_i/2) \geq cT_i$, i.e., to $L_i(\lambda) = q_i e^{-W_0(\cdot)} \leq 1/a_i$; the $[\cdot]^+$ in (4) selects the right case. Since every $T_i > 0$, each q_i is finite and positive; $L_i(\lambda)$ is decreasing in λ because $W_0(z)/z$ is decreasing, so each $P_i(\lambda)$ is continuous and nonincreasing, strictly decreasing while active, with $S(0^+) = \sum_i \bar{P}_i > P_{\text{tot}}$ and $S(\lambda) = 0$ for $\lambda \geq \max_i 2a_i T_i/c$; hence λ^* is unique. $\square$

Waterfilling has the same form, $P_i = [\mu - 1/a_i]^+$, with one water level μ shared by all channels. Here each channel has its own level L_i , set by its $q_i = 2^{T_i}/a_i$. The proof also gives the shortfall directly.

Corollary 1 (Shortfall and water level). *At the optimum, every channel's shortfall is*

$$\delta_i(\lambda^*) = \min \left\{ T_i, \frac{1}{c} W_0\left(\frac{\lambda^* c^2 q_i}{2}\right) \right\}, \quad (6)$$

and every active channel satisfies

$$P_i^* + \frac{1}{a_i} = L_i(\lambda^*) = \frac{2}{\lambda^* c} \delta_i. \quad (7)$$

The water level of each active channel is thus proportional to its own shortfall: channels with larger shortfalls sit higher. Equivalently, the optimum equalizes the shortfall-weighted marginal rate, $2\delta_i \partial r_i / \partial P_i = \lambda^*$.

For intuition, W_0 is increasing, $W_0(z) \approx z$ as $z \rightarrow 0$, and $W_0(z) = \ln z - \ln \ln z + o(1)$ as $z \rightarrow \infty$. When power is plentiful ($\lambda \rightarrow 0$), $\delta_i \approx \lambda c q_i/2$ and $L_i \rightarrow q_i$,

recovering the caps; as power becomes scarce, shortfalls grow only logarithmically in λq_i until they reach T_i and the channel is switched off.

Corollary 2 (Ordering and activity). (i) An active channel's shortfall depends on (a_i, T_i) only through $q_i = 2^{T_i}/a_i$ and increases with it: active channels that need more power to reach their target bear more of the deficit. (ii) Channel i receives zero power iff $a_i T_i \leq c\lambda^*/2$, so channels are switched off in increasing order of $a_i T_i$. Unlike waterfilling, a high target can keep a weak channel active. (iii) Every shortfall is nonincreasing in P_{tot} , and strictly decreasing for active channels: no channel loses from a larger budget.

Proof. (i) and (ii) follow from (6) and the monotonicity of W_0 . For (iii), $S(\lambda)$ is decreasing, so λ^* decreases as P_{tot} grows, and each $\delta_i(\lambda)$ is nondecreasing in λ . $\square$

Remark 1 (The ℓ_p family). Replace the squared penalty by $J_p = \sum_i [(T_i - r_i)^+]^p$. For every $p \geq 1$, J_p is strictly convex on $\mathcal{B}$. For $p > 1$, repeating the proof with $m = p - 1$ gives $\delta_i^m e^{c\delta_i} = \lambda c q_i / p$, hence

$$\delta_i(\lambda) = \min \left\{ T_i, \frac{m}{c} W_0 \left(\frac{c}{m} \left(\frac{\lambda c q_i}{p} \right)^{1/m} \right) \right\}, \quad (8)$$

and again $P_i^* = [L_i - 1/a_i]^+$, now with level $L_i = p \delta_i^{p-1} / (c\lambda) \propto \delta_i^{p-1}$. At the endpoint $p = 1$, $J_1 = \sum_i T_i - \sum_i \min(r_i, T_i)$ on $\mathcal{B}$, and the KKT conditions give a common level $P_i + 1/a_i = \mu$ capped at q_i : capped waterfilling. As $p \rightarrow \infty$ the objective approaches minimax shortfall, $\min \max_i \delta_i$. The exponent p is thus a single knob from served demand to equity, with the closed form (8) at every interior point.

Remark 2 (Extensions). Weights $w_i > 0$ in J replace λ by λ/w_i in (4) and (6), and the bracket of Algorithm 1 becomes $\lambda_{\text{lo}} = (\sum_i \bar{P}_i - P_{\text{tot}}) / (c^2 \sum_i q_i^2 / w_i)$, $\lambda_{\text{hi}} = \max_i 2w_i a_i T_i / c$; $w_i = T_i^{-2}$ penalizes relative rather than absolute shortfall. A coding scheme with SNR gap Γ , $r_i = \log_2(1 + a_i P_i / \Gamma)$, only rescales $a_i \rightarrow a_i / \Gamma$. With separate power budgets for disjoint groups of channels, the problem separates across groups, and each is solved by (4).

V. ALGORITHM

Theorem 2 reduces the problem to finding the root of the monotone scalar function $S(\lambda)$. The following bracket removes any initial search.

Proposition 2 (Explicit bracket). In regime (B), $\lambda^* \in [\lambda_{\text{lo}}, \lambda_{\text{hi}}]$ with

$$\lambda_{\text{lo}} = \frac{\sum_i \bar{P}_i - P_{\text{tot}}}{c^2 \sum_i q_i^2}, \quad \lambda_{\text{hi}} = \max_i \frac{2a_i T_i}{c}. \quad (9)$$

Proof. By the inactivity condition in the proof of Theorem 2, every channel is off at λ_{hi} , so $S(\lambda_{\text{hi}}) = 0 < P_{\text{tot}}$ (assuming $P_{\text{tot}} > 0$). For the lower end, with $\delta_i(\lambda)$ as in (6), $W_0(z) \leq z$ gives $\delta_i \leq \lambda c q_i / 2$, and $1 - 2^{-\delta} \leq c\delta$ gives $P_i(\lambda) = \bar{P}_i - q_i(1 - 2^{-\delta_i}) \geq \bar{P}_i - \lambda c^2 q_i^2 / 2$. Summing, $S(\lambda_{\text{lo}}) \geq \sum_i \bar{P}_i - \frac{1}{2}(\sum_i \bar{P}_i - P_{\text{tot}}) > P_{\text{tot}}$. $\square$

Algorithm 1 Target-rate least-squares allocation

Require: gains a_i , targets $T_i > 0$, budget $P_{\text{tot}} > 0$, tolerance

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 $\varepsilon$ 
1:  $\bar{P}_i \leftarrow (2^{T_i} - 1)/a_i$ ,  $q_i \leftarrow 2^{T_i}/a_i$ 
2: if  $\sum_i \bar{P}_i \leq P_{\text{tot}}$  then return  $\mathbf{P}^* = \bar{\mathbf{P}}$   $\triangleright$  regime (A)
3:  $[\lambda_{\text{lo}}, \lambda_{\text{hi}}] \leftarrow$  bracket (9)
4: while  $\lambda_{\text{hi}}/\lambda_{\text{lo}} > 1 + \varepsilon$  do
5:    $\lambda \leftarrow \sqrt{\lambda_{\text{lo}} \lambda_{\text{hi}}}$   $\triangleright$  bisection on  $\log \lambda$ 
6:    $P_i \leftarrow [L_i(\lambda) - 1/a_i]^+$  via (4), for all  $i$ 
7:   if  $\sum_i P_i > P_{\text{tot}}$  then  $\lambda_{\text{lo}} \leftarrow \lambda$  else  $\lambda_{\text{hi}} \leftarrow \lambda$ 
8: return  $P_i^* = [L_i(\lambda_{\text{hi}}) - 1/a_i]^+$ 

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Each iteration costs N evaluations of W_0 , which Halley's iteration computes to machine precision in a few steps [13]. Bisection on $\log \lambda$ needs $\lceil \log_2(\ln(\lambda_{\text{hi}}/\lambda_{\text{lo}})/\varepsilon) \rceil$ iterations, so the total cost is $O(N(\log(1/\varepsilon) + \log \log(\lambda_{\text{hi}}/\lambda_{\text{lo}})))$; the second term grows only as P_{tot} approaches $\sum_i \bar{P}_i$.

Newton variant. With $u = \ln \lambda$ and $\omega_i = W_0(\lambda c^2 q_i / 2)$,

$$\frac{dS}{du} = - \sum_{i \text{ active}} \frac{(1 + a_i P_i) \omega_i}{a_i(1 + \omega_i)}, \quad (10)$$

which is available in closed form between the points where channels switch off, where S has kinks. We therefore take a Newton step in u , fall back to bisection whenever the step leaves the current bracket, update the bracket at every iteration, and stop when $|S(\lambda) - P_{\text{tot}}| \leq \varepsilon P_{\text{tot}}$.

VI. NUMERICAL RESULTS

We compare against five baselines, each with its own optimality criterion: *waterfilling* ($\max \sum_i r_i$); *capped waterfilling* ($\max$ served demand $\sum_i \min(r_i, T_i)$, the $p = 1$ endpoint of Remark 1); *max-min satisfaction* ($\max \min_i r_i / T_i$, which with equal targets is the $p \rightarrow \infty$ endpoint); *proportional fairness* ($\max \sum_i \log r_i$); and *uniform power*. We checked Algorithm 1, the ℓ_p form (8) (p from 1.25 to 10) and the weighted form against CVXPY/Clarabel on more than 850 random instances; all relative objective gaps are below 10^{-7} , the solver's tolerance. SLSQP agrees on every instance where it was run. Code and data are available at <https://github.com/ANRGUSC/target-rate>.

A. Worked example

Fig. 1 uses $N = 8$ channels with gains $\{20, 15, 10, 7, 5, 3, 2, 1\}$, $T = 3$ b/s/Hz and $P_{\text{tot}} = 10 < \sum_i \bar{P}_i = 16.75$ (regime B). Waterfilling overshoots the four strongest channels by up to 1.95 b/s/Hz and leaves the weakest at 0.63. Capped waterfilling meets five targets exactly and pushes the deficit onto the three weakest channels (rates 2.98, 2.40 and 1.40). The target-rate allocation keeps every channel below but near its target, with shortfalls from 0.09 to 0.98 b/s/Hz ordered by q_i (Corollary 2); its deviation is $J^* = 1.79$, versus 2.92 for capped waterfilling, 3.15 for max-min and 15.4 for waterfilling. Fig. 1(b) shows

Waterfilling Capped waterfilling Max-min satisfaction Target-rate LS

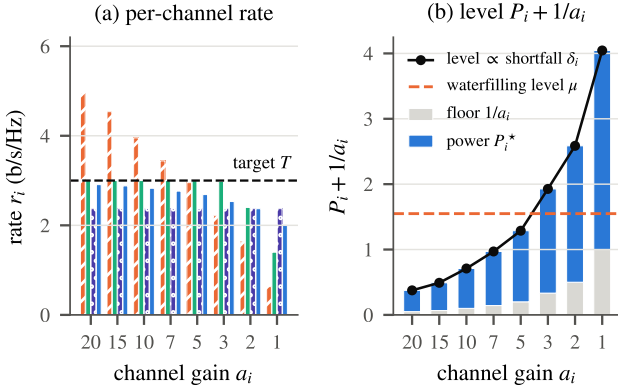


Fig. 1. Worked example ($N = 8$, $T = 3$, $P_{\text{tot}} = 10$). (a) Achieved rates. (b) Target-rate levels $P_i^* + 1/a_i$, proportional to each channel's shortfall, versus the flat waterfilling level.

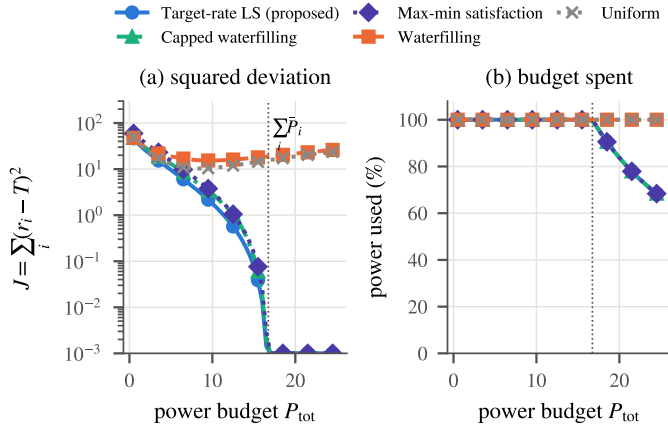


Fig. 2. Budget sweep for the example of Fig. 1. The dotted line marks $\sum_i \bar{P}_i$, the transition from regime (B) to regime (A). Capped waterfilling and max-min also stop at the caps, so these three curves coincide in (b).

the levels $P_i^* + 1/a_i$ of (7), which rise with the shortfall, next to the flat waterfilling level.

Fig. 2 sweeps the budget. Below $\sum_i \bar{P}_i = 16.75$, all methods spend the whole budget and the target-rate allocation has the smallest deviation J . Above it, the allocation reaches $J = 0$ and stops spending: at $P_{\text{tot}} = 25$ it uses 67% of the budget, whereas waterfilling and uniform allocation spend all of it and *increase* J through overshoot. With heterogeneous targets $\{5, 4, 3, 3, 2, 2, 1, 1\}$ ($\sum_i \bar{P}_i = 7.35$), a budget of 15 leaves 7.65 units unused with every target met.

B. Rayleigh-faded channels

We next draw $N = 64$ channels with $a_i = \bar{\gamma} |h_i|^2$, $|h_i|^2 \sim \text{Exp}(1)$ i.i.d., set $P_{\text{tot}} = N$ so that $\bar{\gamma}$ is the mean SNR, use a common target $T = 2$ b/s/Hz, and average 500 realizations per SNR. We report the *satisfaction* $\min(r_i, T)/T$ of each channel. Its mean is the fraction of demand served, which capped waterfilling maximizes by construction; its lower tail (5th percentile, P5) and the fraction of *starved* (zero-power)

TABLE I
RAYLEIGH-FADED CHANNELS ($N = 64$, $T = 2$). SATISFACTION $= \min(r_i, T)/T$; MEAN, P5 (5TH PERCENTILE) AND JAIN INDEX ARE PER REALIZATION, AVERAGED.

Method	$\bar{\gamma} = 5$ dB				$\bar{\gamma} = 10$ dB			
	Mean %	P5 %	Starved %	Jain	Mean %	P5 %	Starved %	Jain
Target-rate	73.9	18.4	3.87	0.90	96.2	85.4	0.18	0.99
Capped WF	79.5	1.0	10.08	0.84	97.8	90.4	0.74	0.98
Max-min	32.1	32.1	0.00	1.00	73.6	73.6	0.00	1.00
Prop. fair	72.8	17.7	0.00	0.86	92.2	49.9	0.00	0.96
Waterfilling	67.7	0.0	16.93	0.74	85.1	6.6	7.03	0.88
Uniform	71.7	13.0	0.00	0.84	89.5	36.5	0.00	0.94

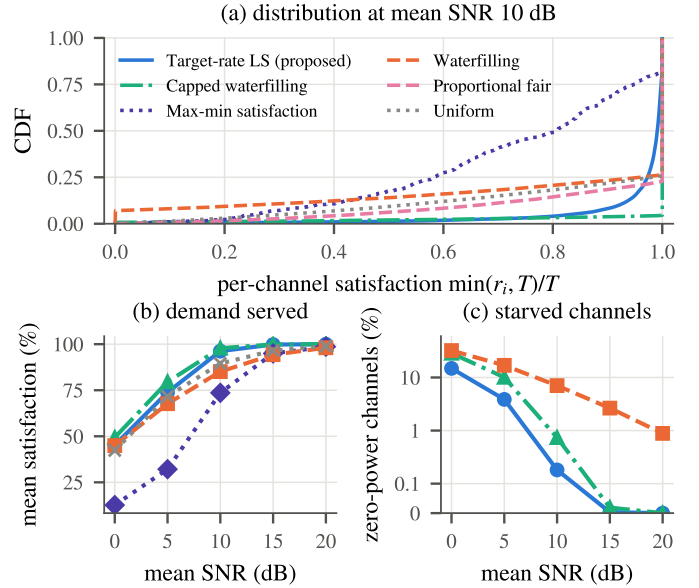


Fig. 3. Rayleigh-faded channels. (a) CDF of per-channel satisfaction at 10 dB. (b) Mean satisfaction (served demand). (c) Fraction of channels given zero power (symmetric-log axis); max-min, proportional fairness and uniform allocation starve none.

channels capture how the deficit is shared among receivers. Metrics are computed per realization and averaged. Table I and Fig. 3 summarize.

Three observations stand out. First, the target-rate allocation gives up little efficiency: it serves 73.9% of demand at 5 dB and 96.2% at 10 dB, versus 79.5% and 97.8% for capped waterfilling, which is optimal for this metric. Second, it spreads the deficit rather than concentrating it. At 5 dB, starved channels fall from 10.1% to 3.9% and the 5th-percentile satisfaction rises from 1.0% to 18.4%. At 10 dB, where only a few deeply faded channels are at risk, capped waterfilling has the better 5th percentile (90.4% vs. 85.4%) but starves $4.1\times$ more channels (0.74% vs. 0.18%). Third, the allocations that never starve behave differently. Max-min serves only 32.1% at 5 dB. Proportional fairness nearly matches the target-rate allocation at 5 dB (72.8% served, P5 17.7%, no starvation), but drops to 92.2% served and a 49.9% P5 at 10 dB, because it keeps spending power above target on strong channels. From

TABLE II
MIXED TARGETS, $T_i \sim \text{Unif}\{1, 2, 4\}$ ($N = 64$). SAT. = MEAN $\min(r_i, T_i)/T_i$; BITS = $\sum_i \min(r_i, T_i)/\sum_i T_i$; ALL IN %.

Method	$\bar{\gamma} = 5$ dB				$\bar{\gamma} = 10$ dB			
	Sat.	Bits	P5	Starved	Sat.	Bits	P5	Starved
Target-rate	59.8	60.5	2.4	8.19	87.0	86.4	55.4	1.05
Target-rate, $w_i = T_i^{-2}$	70.8	62.3	13.1	4.09	90.8	87.1	63.8	0.45
Capped WF	72.8	67.1	0.4	11.29	92.9	90.8	49.7	2.20
Max-min	26.7	26.7	26.7	0.00	57.9	57.9	57.9	0.00
Prop. fair	67.7	57.0	14.4	0.00	86.2	79.6	36.9	0.00
Waterfilling	63.3	56.5	0.0	16.92	81.3	76.5	5.2	7.07
Uniform	67.2	57.6	11.0	0.00	84.8	78.8	28.7	0.00

15 dB on, regime (A) dominates (84% of realizations at 15 dB, 96% at 20 dB): the target-rate allocation meets essentially all demand using 57% and 23% of the budget, while waterfilling, proportional fairness and uniform allocation spend all of it and still serve less.

C. Mixed targets

We now draw each channel's target uniformly from $\{1, 2, 4\}$ b/s/Hz, keeping everything else fixed (Table II). The unweighted objective measures shortfall in absolute bits, and by Corollary 2(ii) channels are switched off in increasing order of $a_i T_i$, so low-target channels tend to be dropped first. At 5 dB it starves 13.6% of the 1-b/s/Hz channels but only 3.9% of the 4-b/s/Hz ones, and its served demand falls to 59.8%. Normalizing the deviation with $w_i = T_i^{-2}$ (Remark 2), i.e., minimizing $\sum_i ((r_i - T_i)/T_i)^2$, restores the equal-target behavior at no extra cost. At 10 dB it serves 90.8% (capped waterfilling: 92.9%), starves 0.45% of channels instead of 2.20%, and has the best P5 of all methods (63.8% vs. 49.7%). Capped waterfilling still delivers the most bits, by construction.

D. The ℓ_p trade-off

Fig. 4 traces the ℓ_p family of Remark 1 on the equal-target channels; every point with $1 < p < \infty$ comes from the closed form (8), and the endpoints from capped waterfilling and minimax bisection. Raising p trades mean satisfaction for fewer starved channels. At 5 dB, moving from $p = 1$ to $p = 2$ costs 5.7 points of served demand and removes 62% of starved channels; continuing to $p = 5$ removes most of the rest but costs another 14.6 points. At 10 dB, $p = 2$ gives up 1.6 points and cuts starvation by 75%. The least-squares objective thus sits near the knee in both regimes, and (8) offers the other operating points at the same cost.

E. Computation time

Table III reports median times over 10 random regime-(B) instances per N , all measured on the same machine. Algorithm 1 is timed compiled (Numba) with bisection ($\varepsilon = 10^{-12}$, 43–44 iterations) and with the safeguarded Newton step (4–6 iterations), and as vectorized NumPy. For the Clarabel interior-point solver, the CVXPY problem is compiled once and we count only the solve time Clarabel reports; SLSQP uses analytic gradients. The Newton variant is 43–103 $\times$ faster than

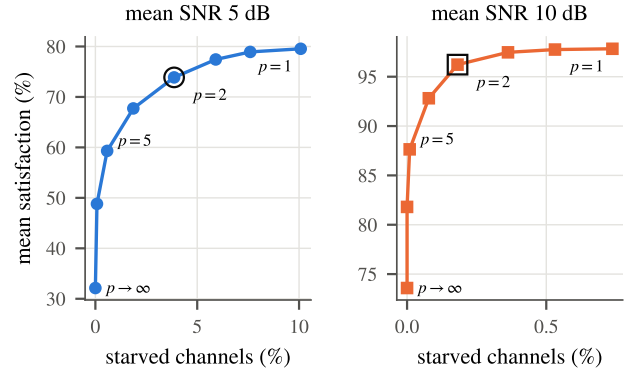


Fig. 4. Served demand versus starved channels across the ℓ_p shortfall family, $p \in \{1, 1.25, 1.5, 2, 3, 5, 10, \infty\}$. The ringed point is $p = 2$.

TABLE III
MEDIAN SOLVE TIME (MS) AND SPEEDUP OF THE NEWTON VARIANT OVER CLARABEL (REGIME B, $a_i = 10 |h_i|^2$, $T = 2$, $P_{\text{tot}} = \min(N, \frac{1}{2} \sum_i \bar{P}_i)$).

N	Algorithm 1			Reference solvers		
	Newton	Bisect.	NumPy	Clarabel	SLSQP	Speedup
8	0.007	0.038	0.45	0.30	2.01	43 $\times$
64	0.043	0.26	1.03	2.30	17.9	54 $\times$
256	0.17	1.03	2.63	12.1	1,006	73 $\times$
1,024	0.66	4.05	8.70	68.1	49,293	103 $\times$

Clarabel's solve time alone, the gap widening with N , and the two agree to a relative objective of 1.3×10^{-9} . At $N = 4096$, Newton takes 2.7 ms, while Clarabel failed on one of three instances and returned an inaccurate solution on another. A practical tolerance $\varepsilon = 10^{-6}$ cuts bisection to 23–24 iterations with objective within 10^{-6} (relative) of the optimum.

VII. CONCLUSION

Minimizing the squared deviation from per-channel rate targets gives an allocation that never overshoots, returns surplus power, and has a closed form: the optimal shortfall of each channel is a Lambert-W function of $2^{T_i}/a_i$ and a scalar dual variable. The same form covers every ℓ_p penalty with $1 < p < \infty$, placing capped waterfilling and minimax allocation at the two ends of one family. The optimum costs a few Lambert-W evaluations per channel. Open directions include imperfect CSI, discrete rate sets, finite-blocklength rate models, and joint channel assignment with the closed form as the inner solver.

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